Ajinkya Pawar

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EDUCATION

University of Michigan, Ann Arbor [2026 – 2028]
Master of Science in Robotics | Coursework: Self Driving Cars, Mathematics for Robotics, Robotic Systems Lab

Indian Institute of Technology Bombay [2022 – 2026]
B.Tech in Metallurgical Engineering & Materials Science, Minor in Artificial Intelligence & Data Science *CGPA: 8.41/10*

AUTONOMOUS RACING EXPERIENCE

Led the Level 4 autonomous vehicle division in a team of 100+ *students representing* ***India*** *as the* 1st *and the only Indian self-driving EV racecar contingent among* 80+ *teams at the Formula Student competitions internationally*

Deputy Team Lead & Chief of Autonomous Systems | *IIT Bombay Racing Driverless* [Mar 2023 – Jul 2026]

- Won 1st overall at Formula Student Portugal 2026 (Driverless Cup, EV) and 1st in the Engineering Design Presentation
- Secured 4th place among 25 teams in the Formula Student AI 2025 competition, a 7-place jump from previous year
- Developed a cutting-edge LiDAR-camera fusion pipeline for enhanced perception performance in the dynamic events
- Reduced end-to-end latency from 85 ms to 18 ms by eliminating CPU-GPU copies, multithreading, and concurrency
- Optimized YOLOv11 detection to 3.2 ms/frame via TensorRT inference on Jetson, an 8.7x speedup over PyTorch
- Built a five-stage LiDAR-only pipeline (Patchwork++, DBSCAN, PCA, intensity filtering, ONNX classification)
- Built a ROS-integrated stack on Linux with sensor fusion of LiDAR, camera, and IMU for robust state estimation
- Designed and implemented an NMPC using a dynamic bicycle model to minimize lap time with a 15% performance gain
- Cut cross-track error 16% by tuning nonlinear pure pursuit; built a Delaunay and RRT* planner for dynamic tracks

RESEARCH EXPERIENCE

Robotics Research Intern | *ARC Lab, Rutgers University (Prof. Jingjin Yu)* [Apr 2025 – Jul 2025]

- Clutter removal: replaced the MLP-based push interaction predictor with a Transformer model, improving accuracy by over 40% and enabling robust generalization to diverse cluttered scenes with variable object configurations
- Developed a baseline spiral push policy, validated in Isaac Gym simulations and real-world 6-DoF UR5e experiments
- Ran hyperparameter tuning and ablation studies on architecture choices; explored end-to-end RL push policies

Contact Force Control of Aerial Manipulators | *Undergraduate Thesis, IIT Bombay* [Aug 2025 – May 2026]

- Designed a cascaded PID controller for a quadcopter with an under-actuated manipulator (under 2% steady-state error)
- Implemented spline-based 3D path planning, used VICON for indoor localization, and worked on contact force control

PROJECTS

Vision-Guided Robot Arm Manipulation (Armlab) | *ROB 550 Robotic Systems Lab, UMich* [Sep 2026 – Present]

- Developed the block detector: tag-homography top-down warp, plane-fit depth flattening, and height thresholding
- Extracted block centroids (image moments), orientation (minAreaRect), and HSV color and size labels from contours
- Calibrated extrinsics from AprilTags (solvePnP) and mapped pixel plus depth to world coordinates for pick locations
- Implemented Lite 6 forward kinematics (DH, product of exponentials) and geometric plus damped Gauss-Newton IK
- Integrated detection and inverse kinematics into a click-to-pick-and-place state machine on the 6-DOF arm

Autonomous Mobile Robot Navigation (Botlab) | *ROB 550 Robotic Systems Lab, UMich* [Sep 2026 – Present]

- Built LiDAR occupancy-grid mapping (deskewing, Bresenham, log-odds) and particle-filter localization for SLAM
- Calibrated the MBot camera for rectified-image AprilTag detection and implemented A* planning for maze navigation
- Tuned PID velocity control, implemented gyrodometry, and wrote a rotate-translate-rotate path-tracking controller

AWARDS AND PUBLICATIONS

- IIT Bombay Undergraduate Research Award (2025) for perception and model predictive control in autonomous racing
- K. Boyalakuntla, A. Pawar, et al., "Plan-Conditioned Imitation for Robust Object Retrieval under Self-Occlusion in Dense Clutter," ICRA 2027 (under review) (arXiv:2609.38857)
- Y. Rampuria et al., "IIT Bombay Racing Driverless: Autonomous Driving Stack for Formula Student AI," contributing author on perception, path planning, and controls (arXiv:2408.06113)

TECHNICAL SKILLS

Perception & Learning: Object Detection, Segmentation, Camera Calibration, LiDAR, Sensor Fusion, SLAM

Control & Planning: NMPC, Cascaded PID, Pure Pursuit, RRT*, A*, Inverse Kinematics, State Estimation

Languages & Tools: Python, C++, C, CUDA, PyTorch, TensorRT, OpenCV, CasADi, ROS, CARLA, Gazebo, MATLAB